Analytical chemistry 1 for engineers

Homework 10 – Gran plot

姓名： **Name:** **ID number:** **Major:**

H10-1. The acid-base titrations on an unidentified monoprotic acid solution. Titration of 100 mL of the unknown acid with 0.009141 M NaOH.

V (NaOH, mL)	pH	V (NaOH, mL)	pH
0.00	4.54	37.34	6.22
1.01	4.55	37.59	6.25
2.20	4.61	37.89	6.29
3.29	4.70	38.09	6.34
4.19	4.72	38.35	6.39
5.17	4.79	38.61	6.48
6.23	4.81	38.90	6.53
7.40	4.88	39.14	6.65
8.58	4.90	39.35	6.71
9.63	4.97	39.51	6.80
10.74	5.00	39.75	6.98
11.50	5.02	40.00	7.30
12.90	5.04	40.21	8.60
13.99	5.11	40.30	9.12
15.10	5.15	40.53	9.43
16.15	5.20	40.79	9.64
17.30	5.22	40.91	9.71
18.49	5.26	41.11	9.80
19.78	5.30	41.30	9.88
20.83	5.33	41.59	9.96
21.97	5.38	41.80	10.00
22.85	5.40	42.10	10.06
23.89	5.42	42.29	10.10
25.02	5.49	42.51	10.15
26.11	5.50	42.75	10.20
27.21	5.54	43.00	10.21
28.09	5.60	44.01	10.29
29.29	5.63	45.10	10.38
30.51	5.69	46.25	10.43
31.50	5.75	47.33	10.49
32.70	5.80	48.39	10.50
33.65	5.89	49.49	10.52
34.83	5.95	50.69	10.59
35.73	6.02	51.89	10.60
36.89	6.15	52.89	10.61
37.11	6.21	54.00	10.62

The data collected from the titration experiment are given in the table. Please try to find the end point Ve and *K*a of this weak acid by use of the methods of first-derivative and Gran plot, respectively.

First-derivative method:

V (NaOH, mL)	pH	Average V $\bar{x} = \frac{x_1 + x_2}{2}$	Slope $k = \frac{y_2 - y_1}{x_2 - x_1}$
38.35	6.39	38.48	0.346154
38.90	6.53	38.755	0.172414
39.14	6.65	39.02	0.5
39.35	6.71	39.245	0.285714
39.51	6.80	39.43	0.5625
39.75	6.98	39.63	0.75
40.00	7.30	39.875	1.28
40.21	8.60	40.105	6.190476
40.30	9.12	40.255	5.777778
40.53	9.43	40.415	1.347826
40.79	9.64	40.66	0.807692
40.91	9.71	40.85	0.583333
41.11	9.80	41.01	0.45

The maximum slope, k = 6.19, and the volume of base is the Ve = 40.105 mL.

$x \times 100$ mL acid = 0.009141 M × 40.105 mL NaOH

x = 0.003666 M

When the Vb = $\frac{1}{2}$Ve, the pH is the pKa.

$\frac{1}{2}$Ve = 20.0525 mL, which is between these two data

19.78	5.30
20.83	5.33

So, at Vb = 20.0525 mL,

pH = pKa = 5.30 + $\frac{20.0525-19.78}{20.83-19.78}$×(5.33-5.30) = 5.31

$K_a = 10^{-5.31} = 4.90 \times 10^{-6}$

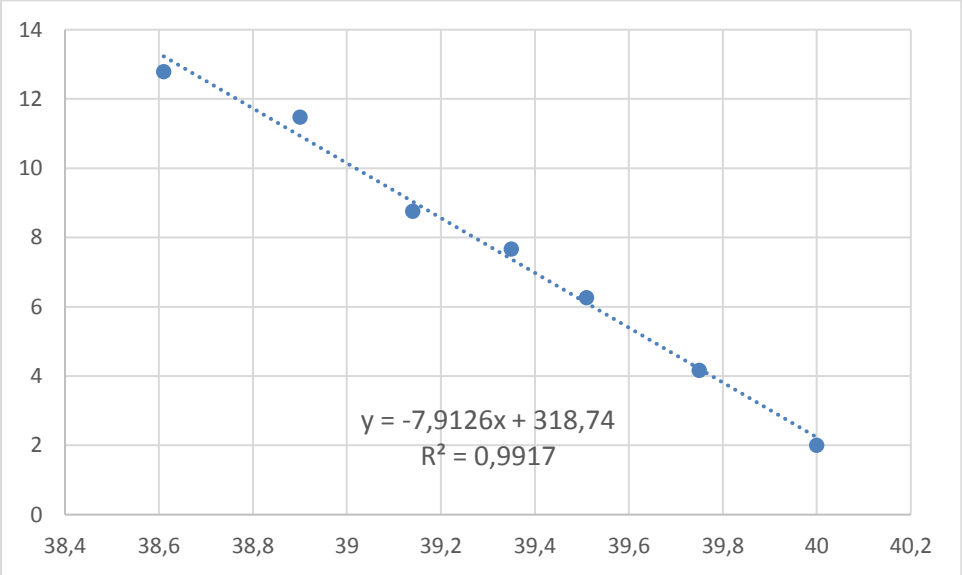
Gran plot method: The data before the equivalence point are employed,

$$V_b \times 10^{-pH} = K_a V_e - K_a V_b$$

Prior to the equivalence point, when weak acid is titrated with a strong base, the Gran function is,

$$V_b \times 10^{-pH} = K_aV_e - K_aV_b$$

x(Vb)	y(pH)	$10^6 \times x \times 10^{-y}$
38.61	6.48	12.78
38.90	6.53	11.48
39.14	6.65	8.76
39.35	6.71	7.67
39.51	6.8	6.26
39.75	6.98	4.16
40.00	7.3	2



Set y = 0, the intercept of x is 40.28 mL

$x \times 100$ mL acid = 0.009141 M × 40.28 mL NaOH

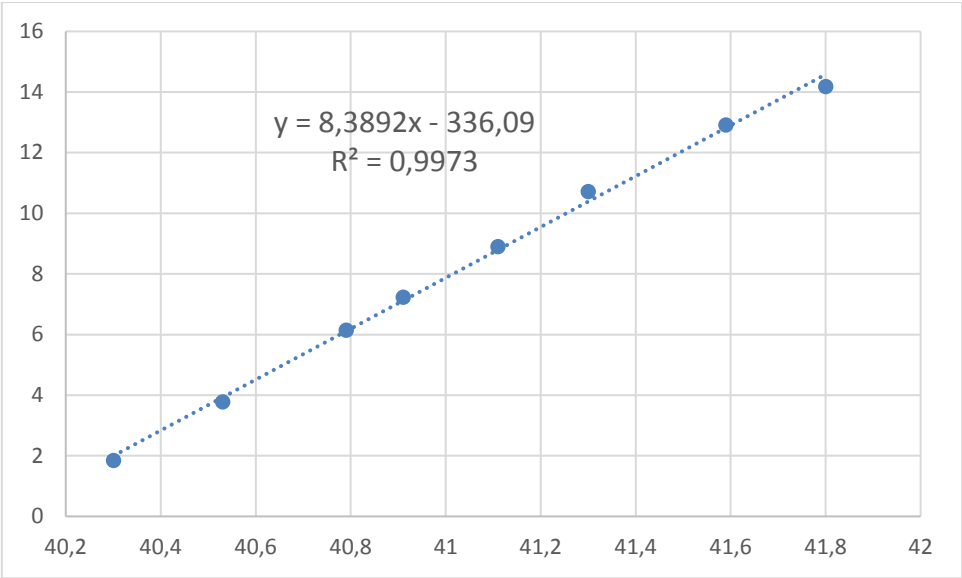
x = 0.003682 M

The Ka is the slope of k. So, Ka = $7.9126\times10^{-6} \approx 7.91 \times10^{-6}$

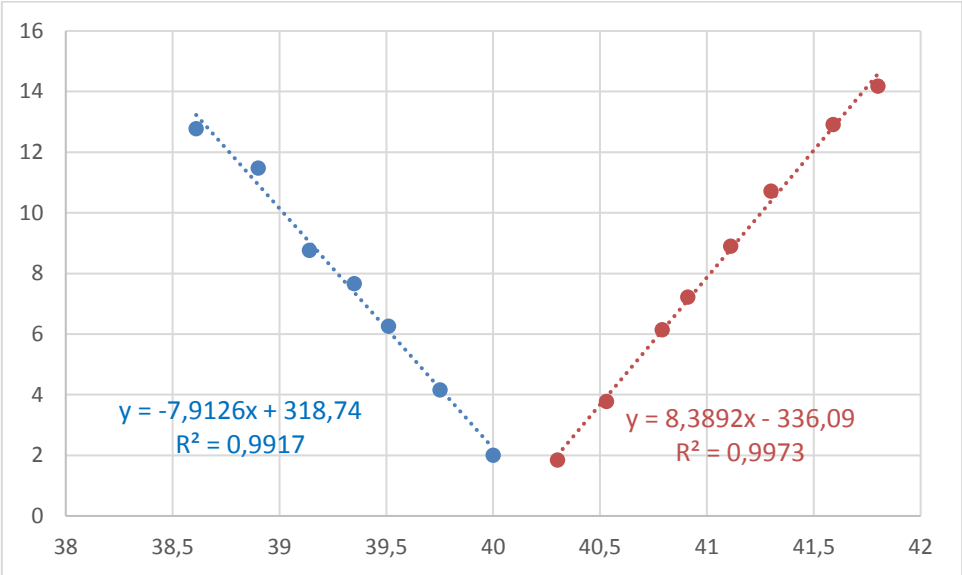
Expansion-1

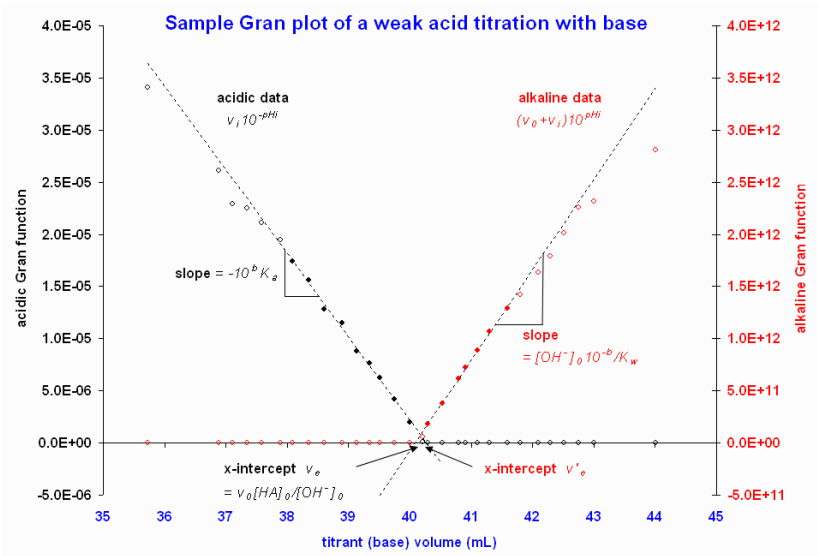
Another side of data (V > Ve) to plot the Gran plot by V_b v.s. $[(V_0 + V_b)\times10^{pH}]$ also can be carried out. Then we can obtain Ve = 40.06 mL as shown in the figure below (This part is just for extension, you can learn more about it if you are interesting).

$x(V_b)$	$y(pH)$	$10^6 \times (x+100) \times 10^y$
40.30	9.12	1.85
40.53	9.43	3.78
40.79	9.64	6.15
40.91	9.71	7.23
41.11	9.80	8.90
41.30	9.88	10.72
41.59	9.96	12.91
41.80	10.00	14.18



The x intercept is 40.06 mL. The formula for this method can be developed by yourselves.





Expansion-2

When weak base B is titrated with a strong acid, the Gran function is,

$V_a \cdot 10^{pH} = \frac{1}{k_a}V_e - \frac{1}{k_a}V_a$

Ka is the acid dissociation constant of BH^+.

The formula for this method can be developed by yourselves if you are interesting on it.

H10-2. In experiment, 10 mL unknown Fe^{2+} solution was titrated with 0.1 M Ce^{4+} as a redox titration, which was monitored by potentiometric method. The results were presented in the table. By plotting the measured potential (with the factor) along the titrant volume by Gran plot, the end point and the concentration of Fe^{2+} can be obtained. The Redox reaction is,

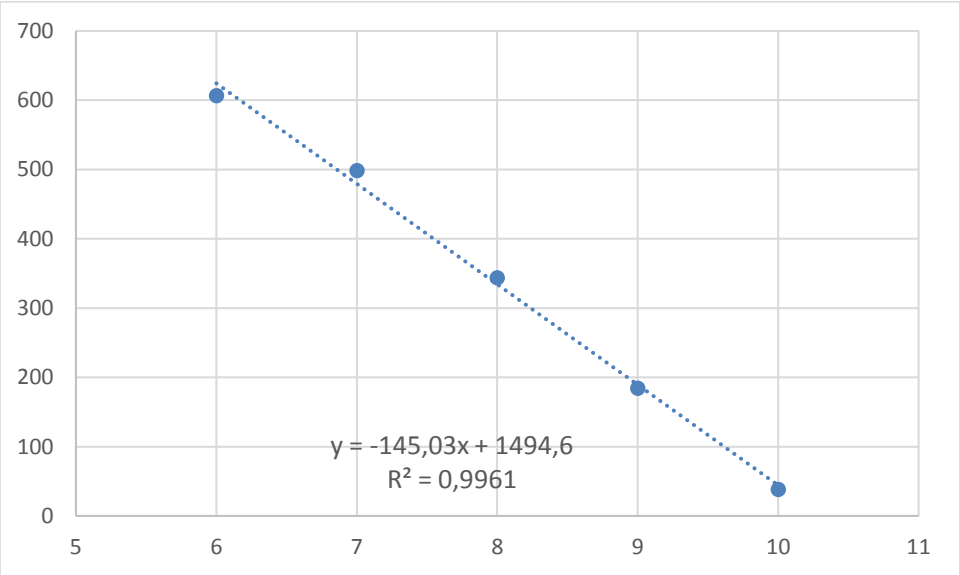
$$Fe^{2+} + Ce^{4+} \rightleftharpoons Fe^{3+} + Ce^{3+}$$

Volume (V) of Ce^{4+} /mL	Potential (E) /V
0	0.382
1	0.416
2	0.431
3	0.443
4	0.453
5	0.462
6	0.473
7	0.482
8	0.495
9	0.514
10	0.557
11	1.154
12	1.179
13	1.193
14	1.202
15	1.209
16	1.214

Firstly, compute the new $\mathbf{Y = 10^{11} \times V \times 10^{-E/0.05916}}$,

Volume (V) of Ce^{4+} / mL	Potential (E) / V	Y ($10^{10} \times V \times 10^{-E/0.05916}$)
0	0.382	0
1	0.416	929.4409
2	0.431	1036.817
3	0.443	974.8854
4	0.453	880.7632
5	0.462	775.6028
6	0.473	606.5745
7	0.482	498.5413
8	0.495	343.5184
9	0.514	184.4759
10	0.557	38.44631
11	1.154	3.43E-09
12	1.179	1.41E-09
13	1.193	8.88E-10
14	1.202	6.74E-10
15	1.209	5.5E-10
16	1.214	4.83E-10

Then, choose the data before the equivalence point of Ve (From 6 mL to 10 mL) and plot the diagram with the V and Y as following,



The *x*-intercept is end point and Ve = 10.31 mL.

So, the $[Fe^{2+}] \times 10$ mL = 0.1 M × 10.31 mL Ce^{4+}

$[Fe^{2+}] = 0.1031$ M ≈ 0.1 M